Ques 1 : Simple Variable Declaration and Output

Declare three variables and print their sum.

Instructions:

1. Declare a variable named num1 using const and assign it the value 15.
2. Declare a variable named num2 using let and assign it the value 25.
3. Declare a variable named total and assign it the arithmetic sum of num1 and num2.
4. Print the value of total to the console.

Sample Input : num1 = 15, num2 = 25

Expected Output : sum of 15 and 25 is 40.

Ques 2 : String Concatenation in console.log()

Demonstrate the difference between addition and concatenation.

Instructions:

1. Declare a variable named item using const and set its value to the string "Phone".
2. Declare a variable named price using let and set its value to the number 50000.
3. Print a single string to the console that says: "The Phone costs 50000 rupees." You must use the item and price variables in the console.log() statement.

Sample input : item = “phone”, price = 50000

Expected Output : The phone cost 50000 rupees

Ques 3 : The Comma Operator in console.log()

Goal: Practice using the comma separator in console.log() to print a variable's label and value clearly.

Instructions:

1. Declare a const variable userRole and set it to "Administrator".
2. Use a single console.log() statement to print the string literal "Role:" followed by the value of the userRole variable, separated by a comma (no string concatenation).

Sample input : userRole = “Administrator”

Expected output : Role : Administrator

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Ques 4 :Re-assigning let

Practice the re-assignment capability of the let keyword.

Instructions:

1. Declare a variable named counter using let and initialize it to 0.
2. Print the initial value of counter.
3. Re-assign the value of counter to be counter + 5.
4. Print the new value of counter.

Sample input : counter = 0;

Expected output : counter value is changed, new value : 5

Ques 5 : Undefined Variable Output

See the output when a variable is declared but not initialized.

Instructions:

1. Declare a variable named data using let but **do not** assign it a value.
2. Print the value of data to the console.
3. Print the **data type** of data using typeof (e.g., console.log(typeof data);).

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Ques 6 : Variable Swapping

Use a temporary variable to swap the values of two variables.

Instructions:

1. Declare let variables: a = 5 and b = 10.
2. Declare a third variable, temp, to temporarily hold one of the values.
3. Write the three lines of code required to swap the values of a and b (so a becomes 10 and b becomes 5).
4. Print the final values of a and b in two separate console.log() statements.

Sample input : a = 5, b = 10

Expected output : before swapping : a = 5, b = 10, after swapping : a = 10, b = 5.

Ques 7 : Printing Multiple Values with Commas

Goal: Use a single console.log() call to output several variables separated by commas.

Instructions:

1. Declare let variables: name = "Sara", age = 28, and city = "Seattle".
2. Use a **single** console.log() statement to print the values of all three variables, separated by commas (without using string concatenation or template literals).

Sample input : name = “sara”, age = 28, city = “Seattle”

Expected output : sara 28 Seattle

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Ques 8 : Calculating Area with Constants

Use constants for fixed mathematical values and calculate a formula.

Instructions:

1. Declare a constant PI with the value 3.14159.
2. Declare a constant radius with the value 5.
3. Calculate the area using the formula $Area = \pi r^2$. (You can use the multiplication operator * twice for r^2: PI * radius * radius).
4. Print the calculated area to the console.

Sample input : pI = 3.14159 , radius = 5

Expected output : 78.53975

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Ques 9 : Variable from User Input Simulation

Simulate taking user input and confirming the data type.

Instructions:

1. Declare a variable named userInput using let and use the built-in browser function prompt("Enter a number:") to assign it a value (when running in a browser environment).
2. Print the value of userInput.
3. Print the **data type** of userInput using typeof (e.g., console.log(typeof userInput);).

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Ques 10 : Hoisting and var (ReferenceError avoidance)

Goal: Observe the effects of var hoisting where a variable is accessed before its declaration, resulting in undefined.

Instructions:

1. Print the value of a variable named hoistedVar to the console.
2. **After** the console.log() statement, declare the variable hoistedVar using **var** and assign it the value "Defined Later".

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